

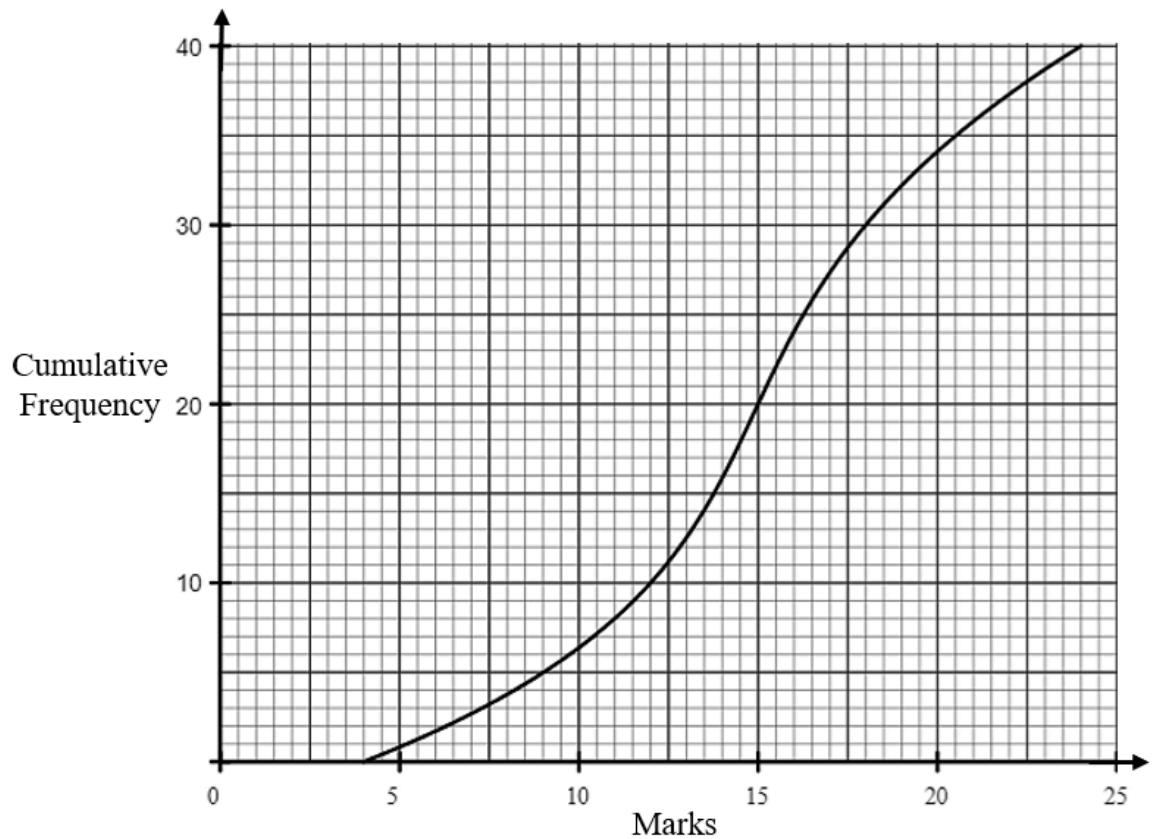
- (c) Calculate, in square metres, the actual area of triangle PQR .

Answer _____ m^2 [3]

- (d) A drone flies along the line QP at a vertical height of 75m.
Find the greatest angle of elevation of the drone from R .

Answer _____ [4]

- 8 The marks attained by 40 students in a Mathematics test were recorded. The cumulative frequency curve shows the distribution of the marks.



- (a) Use the curve to estimate the
(i) the median mark,

Answer _____ [1]

- (ii) the interquartile range.

Answer _____ [2]

- (b) 12.5% of students achieved more than x marks in this test. Estimate the value of x .

Answer $x =$ _____ [1]

- (c) Complete the frequency distribution table of the marks attained by the students.

Marks (x)	$4 \leq x < 10$	$10 \leq x < 15$	$15 \leq x < 20$	$20 \leq x < 24$
Number of students				

[2]

Calculate an estimate for

- (i) the mean mark,

Answer _____ [1]

—

- (ii) the standard deviation of their marks.

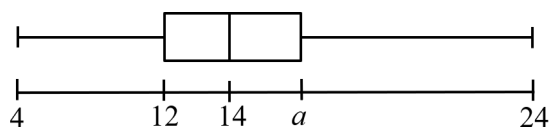
Answer _____ [1]

—

The same group of students sat for a Chemistry test.

The maximum mark for the test was also 25.

The box-and-whisker plot of the distribution of the marks is shown below.



- (d) The scores of the top 25% of the students for the Chemistry test were less consistent than the scores of the bottom 25%.

Given that a is an integer, write down the value of a .

Answer $a =$ _____ [1]

- (e) Make two comparisons between the performances of the students in the Mathematics test and the Chemistry test.

Use figures to support your answer.

1. _____

2. _____